

UI/UX Screen Specification — Communication & Notifications Module

Reuses RGD v2.0, Appendix A-N, and prior UI/UX Screen Specifications by reference only. FR-37 to FR-39, BR-COM-001 to BR-COM-005, Circular/NotificationLog entities (Appendix-G/H) cited by ID. Student, Guardian, Employee, Class, AcademicSession reused unchanged. Notification Templates and Scheduled Notifications reference fields not modeled in Appendix-G/H — flagged inline (*) as Schema Extension items.

#	Screen ID	Screen Name
1	SCR-COM-001	Communication Dashboard
2	SCR-COM-002	Announcement Management
3	SCR-COM-003	Notice Board
4	SCR-COM-004	SMS Management
5	SCR-COM-005	Email Management
6	SCR-COM-006	Notification Templates
7	SCR-COM-007	Scheduled Notifications
8	SCR-COM-008	Communication History
9	SCR-COM-009	Communication Search
10	SCR-COM-010	Communication Settings

SCR-COM-001 — Communication Dashboard

Purpose	Landing summary of recent announcements, dispatch status, and delivery health.
User Roles	Admin Staff, Teacher, IT Admin, Management (read-only)
Navigation Path	Communication > Dashboard
Page Header	"Communication Dashboard"
Breadcrumb	Home > Communication > Dashboard
Form Sections	Recent Announcements, Dispatch Status, Delivery Health, Quick Links
Field Display Order	Recent Announcements (widget) -> Dispatch Status (widget) -> Delivery Health (widget) -> Quick Links
Conditional Visibility Rules	"Delivery Health" widget (per-channel status) shown only to Admin Staff/IT Admin.
UI Behaviour	Widgets refresh on load; no inline editing.
Buttons & Actions	Go to Announcement Management, Go to Communication History, View Reports
Button Placement	Widget-level action links.
Toolbar Actions	N/A
Table Columns	N/A (widget-based)
Default Visible Columns	N/A
Hidden Columns	N/A
Row Actions	N/A
Filters	Academic Session
Default Sorting	N/A
Default Pagination	N/A
Bulk Actions	N/A
Export Formats	N/A
Print Behaviour	N/A
Empty State	"No communication activity yet."
Loading State	Skeleton widgets while data loads.
Popup/Dialog Details	N/A
Audit Information	N/A (summary view).
Responsive Behaviour	Widgets stack vertically on narrow viewports (NFR-MOB-001).
Role Permissions	All listed roles: Read.
Related Business Rules	BR-COM-004 (delivery status visibility)

Success/Error Condition	Message
Load Error	"Unable to load dashboard. Please retry."

SCR-COM-002 — Announcement Management

Purpose	Create, edit, and post circulars/announcements (including homework posts), per FR-37/FR-38.
User Roles	Admin Staff, Teacher, IT Admin
Navigation Path	Communication > Announcement Management
Page Header	"Announcement Management"
Breadcrumb	Home > Communication > Announcements
Form Sections	Announcement List, Add/Edit Announcement
Field Display Order	Post Type -> Title -> Body -> Target Audience -> Post
Conditional Visibility Rules	"Target Audience" restricted to Teacher's own assigned Class/Section when Post Type = Homework and role = Teacher.
UI Behaviour	Auto Save: draft saved every 30 seconds. Confirm Dialog: shown on Post (bulk-audience posts, BR-COM-002). Unsaved Changes Warning: yes.
Buttons & Actions	Save Draft, Post, Retract, Cancel
Button Placement	Bottom action bar.
Toolbar Actions	N/A
Table Columns	Title, Post Type, Target Audience, Posted Date, Status
Default Visible Columns	All visible
Hidden Columns	None
Row Actions	Edit, Retract, View Delivery Status
Filters	Post Type, Target Audience, Date Range
Default Sorting	Posted Date, descending
Default Pagination	25 rows/page
Bulk Actions	N/A
Export Formats	Excel (post log)
Print Behaviour	N/A
Empty State	"No announcements posted yet."
Loading State	Loading indicator on post/save.
Popup/Dialog Details	Bulk-authorization confirm dialog when audience exceeds configured threshold (BR-COM-002).
Audit Information	Post creation, edit, retract logged with actor, timestamp.
Responsive Behaviour	Form stacks single-column on narrow viewports.
Role Permissions	Admin Staff/Teacher: Create, Read, Update (own posts), Retract (own). IT Admin: full authority.
Related Business Rules	BR-COM-002 (bulk-authorization threshold), FR-37, FR-38

Field Name	Type	Mandatory	Validation	Default	Placeholder	Min/Max Len	RO/Editable
Post Type	Dropdown	Yes	Homework/Circular/Announcement	Circular	—	—	Editable
Title	Text	Yes	Non-empty	—	Enter title	3/150	Editable
Body	Text area	Yes	Non-empty; sanitized against XSS (NFR-SEC-002)	—	Enter content	10/2000	Editable
Target Audience	Dropdown/Multi-select	Yes	Valid Class/Section or "All"	All	Select audience	—	Editable

Success/Error Condition	Message
Success	"Announcement posted."
Blocked -- Threshold	"This audience size requires additional authorization." (BR-COM-002)

SCR-COM-003 — Notice Board

Purpose	Read-only display of active circulars/announcements/homework for the Parent/Student/Staff portal, per FR-38.
User Roles	Parent/Guardian, Student, Teacher, Admin Staff (read-only)

Navigation Path	Portal > Notice Board
Page Header	"Notice Board"
Breadcrumb	Home > Notice Board
Form Sections	Notice List
Field Display Order	Post Type Filter -> Notice Cards (Title, Date, Excerpt)
Conditional Visibility Rules	Only posts targeted at the viewer's own Class/Section or "All" are visible (audience scoping).
UI Behaviour	List refreshes on load; "Load More" pagination rather than full reload.
Buttons & Actions	View Full Post, Filter by Type
Button Placement	Filter top; "View Full Post" inline per card.
Toolbar Actions	Post Type Filter Toggle
Table Columns	N/A (card-based list, not tabular)
Default Visible Columns	N/A
Hidden Columns	N/A
Row Actions	View Full Post
Filters	Post Type
Default Sorting	Posted Date, descending (most recent first)
Default Pagination	20 cards per load ("Load More")
Bulk Actions	N/A
Export Formats	N/A (view-only)
Print Behaviour	Individual notice print-ready.
Empty State	"No notices posted yet."
Loading State	Loading indicator/skeleton cards while list loads.
Popup/Dialog Details	Full-post detail opens as a modal dialog.
Audit Information	N/A (read-only view).
Responsive Behaviour	Cards reflow to single-column on narrow viewports.
Role Permissions	Parent/Guardian/Student/Teacher/Admin Staff: Read (scoped to own Class/Section or "All").
Related Business Rules	FR-38 (audience scoping)

Success/Error Condition	Message
Load Error	"Unable to load notice board."

SCR-COM-004 — SMS Management

Purpose	Monitor and manage SMS-channel dispatch activity, per FR-37 (Communication module Section 13, Appendix-I).
User Roles	Admin Staff, IT Admin
Navigation Path	Communication > SMS Management
Page Header	"SMS Management"
Breadcrumb	Home > Communication > SMS Management
Form Sections	SMS Dispatch List
Field Display Order	Date Range -> Status Filter -> Dispatch List
Conditional Visibility Rules	"Retry" action shown only for rows with Status = Failed.
UI Behaviour	List refreshes on filter change; no inline editing (dispatch is event-driven, not manually composed here).
Buttons & Actions	Retry Failed, Export
Button Placement	"Retry" inline per failed row; "Export" top-right.
Toolbar Actions	N/A
Table Columns	Recipient, Trigger Event, Status, Dispatched At
Default Visible Columns	All visible
Hidden Columns	None

Row Actions	Retry, View Detail
Filters	Status, Date Range, Trigger Event
Default Sorting	Dispatched At, descending
Default Pagination	50 rows/page
Bulk Actions	Bulk Retry (all Failed rows in current filter)
Export Formats	Excel, CSV
Print Behaviour	N/A
Empty State	"No SMS dispatch activity for this filter."
Loading State	Loading indicator on filter change/retry.
Popup/Dialog Details	Bulk-retry confirm dialog.
Audit Information	Dispatch and retry events logged via NotificationLog (Appendix-G/H ENT-COM-002).
Responsive Behaviour	Table condenses to essential columns on narrow viewports.
Role Permissions	Admin Staff: Read, Retry. IT Admin: full authority, including gateway-health drill-down.
Related Business Rules	BR-COM-004 (delivery status logging), NFR-MON-002 (independent per-integration monitoring)

Success/Error Condition	Message
Success	"{n} messages retried."
Load Error	"Unable to load SMS dispatch log."

SCR-COM-005 — Email Management

Purpose	Monitor and manage Email-channel dispatch activity, per FR-37 (Communication module Section 13, Appendix-I).
User Roles	Admin Staff, IT Admin
Navigation Path	Communication > Email Management
Page Header	"Email Management"
Breadcrumb	Home > Communication > Email Management
Form Sections	Email Dispatch List
Field Display Order	Date Range -> Status Filter -> Dispatch List
Conditional Visibility Rules	"Retry" action shown only for rows with Status = Failed.
UI Behaviour	List refreshes on filter change; no inline editing.
Buttons & Actions	Retry Failed, Export
Button Placement	"Retry" inline per failed row; "Export" top-right.
Toolbar Actions	N/A
Table Columns	Recipient, Trigger Event, Status, Dispatched At
Default Visible Columns	All visible
Hidden Columns	None
Row Actions	Retry, View Detail
Filters	Status, Date Range, Trigger Event
Default Sorting	Dispatched At, descending
Default Pagination	50 rows/page
Bulk Actions	Bulk Retry (all Failed rows in current filter)
Export Formats	Excel, CSV
Print Behaviour	N/A
Empty State	"No email dispatch activity for this filter."
Loading State	Loading indicator on filter change/retry.
Popup/Dialog Details	Bulk-retry confirm dialog.
Audit Information	Dispatch and retry events logged via NotificationLog (Appendix-G/H ENT-COM-002).
Responsive Behaviour	Table condenses to essential columns on narrow viewports.

Role Permissions	Admin Staff: Read, Retry. IT Admin: full authority, including gateway-health drill-down.
Related Business Rules	BR-COM-004 (delivery status logging), NFR-MON-002 (independent per-integration monitoring)

Success/Error Condition	Message
Success	"{n} messages retried."
Load Error	"Unable to load email dispatch log."

SCR-COM-006 — Notification Templates

Purpose	Maintain reusable message templates for common notification triggers (absence alert, fee reminder, etc.).
User Roles	Admin Staff, IT Admin
Navigation Path	Communication > Notification Templates
Page Header	"Notification Templates"
Breadcrumb	Home > Communication > Templates
Form Sections	Template List, Add/Edit Template
Field Display Order	Template Name -> Trigger Event -> Channel -> Message Body (with placeholders)
Conditional Visibility Rules	Placeholder token list shown contextually based on the selected Trigger Event.
UI Behaviour	Confirm Dialog: shown when editing a template already in active use. Unsaved Changes Warning: yes.
Buttons & Actions	Add Template, Edit, Deactivate, Save
Button Placement	"Add" top-right; "Edit"/"Deactivate" inline per row.
Toolbar Actions	N/A
Table Columns	Template Name, Trigger Event, Channel, Status
Default Visible Columns	All visible
Hidden Columns	None
Row Actions	Edit, Deactivate, Preview
Filters	Trigger Event, Channel, Status
Default Sorting	Template Name, ascending
Default Pagination	25 rows/page
Bulk Actions	N/A
Export Formats	N/A (configuration screen)
Print Behaviour	N/A
Empty State	"No templates configured."
Loading State	Loading indicator on save.
Popup/Dialog Details	Preview dialog rendering sample-filled message.
Audit Information	Template changes logged per Appendix-F NFR-LOG-009.
Responsive Behaviour	Form stacks single-column on narrow viewports.
Role Permissions	Admin Staff: Create, Read, Update. IT Admin: full authority, including Deactivate.
Related Business Rules	* Schema Extension – Client/Product Decision Required. (no Template entity in Appendix-G/H; Circular currently carries title/body directly, not template-driven)

Field Name	Type	Mandatory	Validation	Default	Placeholder	Min/Max Len	RO/Editable
Template Name	Text	Yes*	Non-empty, unique	—	e.g., Absence Alert	3/100	Editable
Trigger Event	Dropdown	Yes*	Valid system event (Admission/Attendance/Fee/Exam)	—	Select trigger	—	Editable
Channel	Dropdown	Yes*	SMS/Email/Push	—	—	—	Editable
Message Body	Text area	Yes*	Non-empty; supports placeholder tokens	—	Enter message with placeholders	10/500	Editable

Success/Error Condition	Message
Success	"Template saved."
Validation Error	"Template name already exists."

SCR-COM-007 — Scheduled Notifications

Purpose	Configure a future dispatch date/time for an announcement or notification rather than immediate posting.
User Roles	Admin Staff, IT Admin
Navigation Path	Announcement Management > Schedule Communication > Scheduled Notifications
Page Header	"Scheduled Notifications"
Breadcrumb	Home > Communication > Scheduled Notifications
Form Sections	Scheduled Queue, Add Scheduled Item
Field Display Order	Announcement/Message -> Scheduled Date -> Scheduled Time -> Recurrence (optional)
Conditional Visibility Rules	"Recurrence" options limited to None for one-off announcements; Daily/Weekly available only for system-configured recurring notice types.
UI Behaviour	Confirm Dialog: shown on scheduling. Unsaved Changes Warning: yes.
Buttons & Actions	Schedule, Cancel Scheduled Item, Edit Schedule
Button Placement	Bottom action bar; "Cancel"/"Edit" inline per queued row.
Toolbar Actions	N/A
Table Columns	Title, Scheduled Date/Time, Recurrence, Status
Default Visible Columns	All visible
Hidden Columns	None
Row Actions	Edit, Cancel
Filters	Status, Date Range
Default Sorting	Scheduled Date, ascending
Default Pagination	25 rows/page
Bulk Actions	N/A
Export Formats	N/A (operational queue screen)
Print Behaviour	N/A
Empty State	"No scheduled notifications."
Loading State	Loading indicator on save.
Popup/Dialog Details	Cancel-scheduled-item confirm dialog.
Audit Information	Scheduling and cancellation events logged with actor, timestamp.
Responsive Behaviour	Table condenses on narrow viewports.
Role Permissions	Admin Staff: Create, Read, Update, Delete (own). IT Admin: full authority.
Related Business Rules	* Schema Extension – Client/Product Decision Required. (no scheduled_for/recurrence field on Circular or NotificationLog in Appendix-G/H; dispatch is currently event-triggered, not time-scheduled)

Field Name	Type	Mandatory	Validation	Default	Placeholder	Min/Max Len	RO/Editable
Scheduled Date	Date	Yes*	Future date	—	—	—	Editable
Scheduled Time	Time	Yes*	Future time if Scheduled Date is today	—	—	—	Editable
Recurrence	Dropdown	No*	None/Daily/Weekly	None	—	—	Editable

Success/Error Condition	Message
Success	"Notification scheduled for {date/time}."
Validation Error	"Scheduled date/time must be in the future."

SCR-COM-008 — Communication History

Purpose	View the complete dispatch history for a specific recipient (student, guardian, or staff) across all channels, per BR-COM-004.
User Roles	Admin Staff, IT Admin; Parent/Guardian, Student, Staff (own, read-only)
Navigation Path	Student Profile > Communication tab Communication > Communication History
Page Header	"Communication History -- {Recipient Name}"
Breadcrumb	Home > Communication > History
Form Sections	Dispatch History List
Field Display Order	Recipient (display) -> Date Range -> Channel Filter -> History List

Conditional Visibility Rules	Recipient scoped automatically to own/linked record for Parent/Guardian/Student/Staff roles.
UI Behaviour	List refreshes on filter change; no inline editing (read-only historical view).
Buttons & Actions	Export, View Message Detail
Button Placement	Top-right action bar; "View Detail" inline per row.
Toolbar Actions	N/A
Table Columns	Date, Channel, Trigger Event, Status
Default Visible Columns	All visible
Hidden Columns	None
Row Actions	View Message Detail
Filters	Channel, Trigger Event, Status, Date Range
Default Sorting	Date, descending (most recent first)
Default Pagination	30 rows/page
Bulk Actions	N/A
Export Formats	Excel, CSV
Print Behaviour	N/A
Empty State	"No communication history found for this recipient."
Loading State	Loading indicator on filter change.
Popup/Dialog Details	Message detail opens as a modal dialog.
Audit Information	N/A (this screen is itself a read view of NotificationLog, which is the audit trail).
Responsive Behaviour	Table becomes stacked cards on narrow viewports.
Role Permissions	Admin Staff/IT Admin: Read (any recipient). Parent/Guardian/Student/Staff: Read (own/linked only).
Related Business Rules	BR-COM-004 (delivery status logging), Appendix-G/H ENT-COM-002 (NotificationLog, reused unchanged)

Success/Error Condition	Message
Load Error	"Unable to load communication history."

SCR-COM-009 — Communication Search

Purpose	Global quick search for an announcement, recipient, or dispatch record, accessible from anywhere in the Communication module.
User Roles	Admin Staff, Teacher, IT Admin
Navigation Path	Global Search Bar (available on all Communication screens)
Page Header	N/A (search widget)
Breadcrumb	N/A
Form Sections	N/A (search widget)
Field Display Order	Search Term -> Results Dropdown
Conditional Visibility Rules	Result set scoped to Teacher's own posts for Teacher role.
UI Behaviour	Loading Indicator: shown while results load (debounced), per NFR-SRCH-001.
Buttons & Actions	Search, Clear
Button Placement	Inline with search field.
Toolbar Actions	N/A
Table Columns	N/A (compact dropdown: Title/Recipient, Type, Date)
Default Visible Columns	N/A
Hidden Columns	N/A
Row Actions	Open Detail
Filters	N/A
Default Sorting	Relevance (best match first)
Default Pagination	Top 10 matches; "View All" links to Communication History
Bulk Actions	N/A

Export Formats	N/A
Print Behaviour	N/A
Empty State	"No matches found."
Loading State	Spinner within search dropdown.
Popup/Dialog Details	N/A
Audit Information	N/A (read-only search).
Responsive Behaviour	Search bar collapses to an icon on narrow viewports, expanding on tap.
Role Permissions	All listed roles: Read (scoped per role).
Related Business Rules	NFR-SRCH-001, NFR-PERF-004

Field Name	Type	Mandatory	Validation	Default	Placeholder	Min/Max Len	RO/Editable
Search Term	Text	Yes (to trigger)	Minimum 2 characters	—	Search by title, recipient, or event	2/100	Editable

Success/Error Condition	Message
No Matches	"No results found for this search term."

SCR-COM-010 — Communication Settings

Purpose	Configure module-wide communication policy values (bulk-audience threshold, default channel preference).
User Roles	IT Admin
Navigation Path	Communication > Settings
Page Header	"Communication Settings"
Breadcrumb	Home > Communication > Settings
Form Sections	Policy Configuration
Field Display Order	Bulk-Audience Authorization Threshold -> Default Channel -> Retry Attempts Limit
Conditional Visibility Rules	N/A (flat configuration form).
UI Behaviour	Confirm Dialog: shown on Save (affects all future dispatch system-wide).
Buttons & Actions	Save Settings, Reset to Default
Button Placement	Bottom action bar.
Toolbar Actions	N/A
Table Columns	N/A (configuration form)
Default Visible Columns	N/A
Hidden Columns	N/A
Row Actions	N/A
Filters	N/A
Default Sorting	N/A
Default Pagination	N/A
Bulk Actions	N/A
Export Formats	N/A
Print Behaviour	N/A
Empty State	N/A
Loading State	Loading indicator on save.
Popup/Dialog Details	N/A
Audit Information	Configuration changes logged per Appendix-F NFR-LOG-009 (Configuration entity, Appendix-G ENT-SYS-005).
Responsive Behaviour	Form stacks single-column on narrow viewports.
Role Permissions	IT Admin: Create, Read, Update.
Related Business Rules	BR-COM-002 (bulk-authorization threshold, configurable)

Field Name	Type	Mandatory	Validation	Default	Placeholder	Min/Max Len	RO/Editable
Bulk-Audience Authorization Threshold	Number	Yes	Positive integer (BR-COM-002)	—	Enter threshold	—	Editable
Default Channel	Dropdown	Yes	SMS/Email/Push	SMS	—	—	Editable
Retry Attempts Limit	Number	Yes	Positive integer	3	—	—	Editable

Success/Error Condition	Message
Success	"Communication settings saved."